



VECTORDB FAQ'S

50 INTERVIEW QUESTIONS & ANSWERS

Q1 What is a Vector Database, and why is it used in AI applications?

A Vector Database stores data as vectors (embeddings) instead of normal text.

It helps AI find similar information based on meaning, not just exact words.

It is mainly used in RAG, chatbots, document search, and recommendation systems.

SHORT ANSWER

"A Vector Database stores embeddings and helps AI quickly find similar information. It is mainly used in RAG and AI applications."

Q2 What is the difference between a traditional SQL database and a Vector Database?

A SQL Database stores text, numbers, and tables. It finds data using exact matches.

A Vector Database stores embeddings (vectors). It finds data based on meaning (similarity).

SHORT ANSWER

"SQL Database finds exact data, while a Vector Database finds similar data based on meaning."

Q3 What is an embedding, and how is it stored in a Vector Database?

An embedding is the numerical representation of data (text, image, audio, etc.).

A Vector Database stores these embeddings as vectors and uses them to find similar data.

SHORT ANSWER

"An embedding is the numerical form of data. A Vector Database stores these vectors and uses them to find similar information."

Q4 How does semantic search differ from keyword search?

Keyword search looks for the exact words.



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9010001847



info@vproskills.com



www.vproskills.com

Semantic search looks for the meaning of the words.

Example:

Keyword: "car" → finds only "car".

Semantic: "car" → can also find "vehicle" or "automobile".

SHORT ANSWER

"Keyword search finds exact words, while semantic search finds data based on meaning."

Q5 What is cosine similarity, and why is it commonly used in Vector Databases?

Cosine similarity measures how similar two vectors are.

The higher the cosine similarity, the more similar the data.

It is commonly used because it compares the meaning of the data, not just the values.

SHORT ANSWER

"Cosine similarity measures how similar two vectors are. It helps the Vector Database find the most relevant information."

Q6 What are Euclidean Distance and Dot Product similarity, and when are they used?

Euclidean Distance measures the distance between two vectors. Smaller distance = More similar.

Dot Product measures how closely two vectors are related. Higher value = More similar.

SHORT ANSWER

"Euclidean Distance compares the distance between vectors, while Dot Product compares their similarity. Both are used to find similar data in a Vector Database."

Q7 What is FAISS, and what are its main features?

FAISS is a Vector Database library developed by Meta. It is used to store vectors and perform fast similarity search.

SHORT ANSWER

"FAISS is a vector search library from Meta. It is fast, efficient, and used to find similar vectors."

Q8 What is ChromaDB, and why is it popular for RAG applications?

ChromaDB is an open-source Vector Database. It stores documents, embeddings, and metadata together.

It is popular because it is simple to use with RAG applications.



SHORT ANSWER

"ChromaDB is an open-source Vector Database. It is popular for RAG because it stores documents, vectors, and metadata together."

Q9 What are the major differences between FAISS and ChromaDB?

FAISS stores only vectors.

ChromaDB stores vectors, documents, and metadata.

SHORT ANSWER

"FAISS is mainly for fast vector search, while ChromaDB also stores documents and metadata, making it better for RAG."

Q10 Can FAISS store metadata along with vectors?

No.

FAISS stores only vectors. Metadata must be stored in another database.

SHORT ANSWER

"No. FAISS stores only vectors. Metadata is stored separately."

Q11 Can ChromaDB store documents, metadata, and embeddings together?

Yes.

ChromaDB can store documents, metadata, IDs, and embeddings in one place.

SHORT ANSWER

"Yes. ChromaDB stores documents, metadata, IDs, and embeddings together."

Q12 Which Vector Database is better for beginners: FAISS or ChromaDB?

ChromaDB is better for beginners because it is easy to use and supports RAG directly.

SHORT ANSWER

"ChromaDB is better for beginners because it is simple and supports documents and metadata."

Q13 How do you create a collection in ChromaDB?

Use the `create_collection()` method.

A collection is like a table in a SQL database.

SHORT ANSWER

"We create a collection using `create_collection()`. It is similar to a table in SQL."

Q14 How do you insert documents into ChromaDB?

Use the `add()` method to insert documents, IDs, metadata, and embeddings.

SHORT ANSWER

"We use the `add()` method to insert documents into ChromaDB."

Q15 How do you perform similarity search in ChromaDB?

Use the `query()` method with a search text or embedding.

ChromaDB returns the most similar documents.

SHORT ANSWER

"We use the `query()` method. ChromaDB returns the most similar documents based on meaning."

Q16 What is an index in FAISS, and why is it important?

An index is a data structure that helps FAISS find similar vectors quickly.

SHORT ANSWER

"An index helps FAISS search vectors faster and improves performance."

Q17 What are the different FAISS index types (Flat, IVF, HNSW, PQ)?

Flat – Exact search

IVF – Faster search for large data

HNSW – Very fast search with high accuracy

PQ – Uses less memory

SHORT ANSWER

"Flat for exact search, IVF and HNSW for fast search, and PQ to save memory."

Q18 What is Approximate Nearest Neighbor (ANN) Search?

ANN finds almost similar vectors very quickly.

It is faster than exact search.

SHORT ANSWER

"ANN finds nearly similar vectors very fast. It is used for large datasets."

Q19 What is Exact Nearest Neighbor Search, and when should it be used?

Exact search finds the most accurate result.

It is best for small datasets.

SHORT ANSWER

"Exact search gives the most accurate result and is suitable for small datasets."

Q20 Which embedding models are commonly used with FAISS and ChromaDB?

Common embedding models are:

OpenAI Embeddings

Sentence Transformers

BGE

E5

SHORT ANSWER

"Common embedding models are OpenAI, Sentence Transformers, BGE, and E5."

Q21 Can you use OpenAI embeddings with FAISS and ChromaDB?

Yes.

Both FAISS and ChromaDB support OpenAI embeddings.

SHORT ANSWER

"Yes. Both FAISS and ChromaDB can use OpenAI embeddings."

Q22 How do you update or delete vectors in ChromaDB?

Use the update() method to update vectors and the delete() method to remove them.

SHORT ANSWER

"Use update() to modify vectors and delete() to remove them."

Q23 Can vectors be updated in FAISS after insertion?

No.

FAISS does not directly support updates. Usually, you delete and rebuild the index.

SHORT ANSWER

"No. FAISS does not support direct updates. Usually, the index is rebuilt."

Q24 How does metadata filtering work in ChromaDB?

Metadata filtering returns only documents that match specific conditions, such as category, author, or date.

SHORT ANSWER

"Metadata filtering searches only the documents that match the given conditions."

Q25 Which Vector Database is more suitable for production environments?

It depends on the project.

FAISS is good for high-speed vector search.

ChromaDB is good for RAG applications.

SHORT ANSWER

"FAISS is best for fast vector search, while ChromaDB is better for RAG applications."

Q26 How do Vector Databases improve Retrieval-Augmented Generation (RAG)?

They retrieve the most relevant documents before sending the question to the LLM.

This improves answer quality.

SHORT ANSWER

"A Vector Database retrieves relevant documents, helping the LLM generate better answers."

Q27 What challenges arise when storing millions of vectors?

The main challenges are:

More memory usage

Slower search

Longer indexing time

SHORT ANSWER

"The main challenges are high memory usage, slower search, and longer indexing time."

Q28 What are the advantages and limitations of ChromaDB?

Advantages:

Easy to use

Stores documents, metadata, and embeddings together

Good for RAG applications

Limitations:

Not as fast as FAISS for very large datasets

Limited scalability compared to enterprise Vector Databases

SHORT ANSWER

"ChromaDB is simple, stores documents and metadata, and is great for RAG. But for very large datasets, FAISS is usually faster."

Q29 How would you choose between FAISS and ChromaDB for a real-world AI project?

Choose FAISS if you need very fast vector search and have a large number of vectors.

Choose ChromaDB if you are building a RAG application and need to store documents, metadata, and embeddings together.

SHORT ANSWER

"I use FAISS for high-speed vector search and ChromaDB for RAG applications because it stores documents, metadata, and embeddings together."

Q30 You have built a RAG application. Why did you choose ChromaDB (or FAISS) over other Vector Databases like Pinecone or Milvus?

I chose ChromaDB because it is easy to use, open-source, and stores documents, metadata, and embeddings together. It is a good choice for small and medium RAG projects.

SHORT ANSWER

"I chose ChromaDB because it is simple, open-source, and works well for RAG applications."

Q31 How do you decide the chunk size and chunk overlap while indexing documents? What problems have you faced with poor chunking?

I usually use 500–1000 characters with 50–100 characters overlap.

Poor chunking can miss important information or return incorrect answers.

SHORT ANSWER

"I use medium-sized chunks with some overlap to keep the context. Poor chunking can reduce answer quality."

Q32 A user says the RAG application is returning irrelevant answers. How would you troubleshoot the issue step by step?

I check:

Embedding model

Chunking

Similarity search

Retrieved documents

Prompt given to the LLM

SHORT ANSWER

"I check the embeddings, chunking, retrieved documents, and prompt to find where the problem is."

Q33 Your application currently uses FAISS. The business now requires metadata filtering (e.g., search only HR documents). How would you redesign the solution?

I would move to ChromaDB or store metadata in a separate database and filter before searching.

SHORT ANSWER

"FAISS does not support metadata filtering directly, so I would use ChromaDB or combine FAISS with a metadata database."

Q34 You have 10 million document embeddings. Which FAISS index would you choose (Flat, IVF, HNSW, PQ) and why?

I would choose HNSW because it provides fast search with good accuracy.

SHORT ANSWER

"For millions of vectors, I would choose HNSW because it is fast and accurate."

Q35 How do you evaluate the quality of an embedding model before using it in a production RAG application?

I test whether it retrieves the correct documents for different user questions.

SHORT ANSWER

"I test the embedding model with real questions and check if it retrieves the correct documents."

Q36 Suppose new documents are added every hour. How would you update the Vector Database without rebuilding the entire index?

I generate embeddings only for the new documents and add them to the existing database.

SHORT ANSWER

"I create embeddings only for new documents and add them to the existing Vector Database."

Q37 What are the biggest performance bottlenecks you've faced while working with Vector Databases, and how did you optimize them?

The main issues were slow search and large memory usage.

I optimized them by using better indexes and proper chunking.

SHORT ANSWER

"Slow search and memory usage were the main issues. I optimized them using better indexes and proper chunking."

Q38 Can you explain the complete lifecycle of a query in your RAG application—from user question to LLM response—including embeddings, similarity search, retrieval, prompt construction, and answer generation?

User asks a question.

Convert the question into an embedding.

Search the Vector Database.

Retrieve the most relevant documents.

Add the documents to the prompt.

Send the prompt to the LLM.

LLM generates the final answer.

SHORT ANSWER

"User question → Embedding → Vector Search → Retrieve Documents → Build Prompt → Send to LLM → Final Answer."

Q39 If you were asked to build an enterprise-scale document search system today, which Vector Database would you choose (FAISS, ChromaDB, Pinecone, Milvus, Weaviate, Qdrant, etc.)? Explain your architecture, technology choices, scalability considerations, and trade-offs.

For an enterprise project, I would choose Pinecone or Milvus because they are built for large-scale production systems.

For small or medium projects, I would use ChromaDB.

SHORT ANSWER

"For enterprise applications, I would choose Pinecone or Milvus because they are highly scalable. For small and medium RAG projects, I would choose ChromaDB because it is simple and easy to manage."

Q40 What is vector dimensionality, and why is it important?

Vector dimensionality is the number of values in a vector.

Higher dimensions can capture more information.

SHORT ANSWER

"Vector dimensionality is the number of values in a vector. It affects the quality of similarity search."

Q41 How are text documents converted into vectors before storing them?

An embedding model converts the text into numerical vectors.

These vectors are stored in the Vector Database.

SHORT ANSWER

"An embedding model converts text into vectors, and those vectors are stored in the Vector Database."

Q42 What is chunking, and why is it necessary for Vector Databases?

Chunking means splitting a large document into smaller parts.

This helps the Vector Database retrieve more accurate information.

SHORT ANSWER

"Chunking splits large documents into small pieces for better search results."

Q43 What is the ideal chunk size for storing documents in a Vector Database?

There is no fixed size.

A common choice is 500–1000 characters with some overlap.

SHORT ANSWER

"A common chunk size is 500–1000 characters with a small overlap."

Q44 How do overlapping chunks improve search accuracy?

Overlap keeps important information that may be split between two chunks.

This improves retrieval quality.

SHORT ANSWER

"Overlap keeps the context between chunks and improves search accuracy."

Q45 What happens if two vectors are very close to each other?

It means the two pieces of data have similar meaning.

SHORT ANSWER

"If two vectors are close, it means the data is semantically similar."

Q46 How does a Vector Database find the nearest neighbors of a query vector?

It compares the query vector with stored vectors using similarity algorithms like Cosine Similarity.

SHORT ANSWER

"It compares the query vector with stored vectors and returns the most similar ones."

Q47 What is similarity threshold, and how does it affect search results?

A similarity threshold is the minimum similarity score required to return a result.

Higher threshold = fewer but more relevant results.

SHORT ANSWER

"The similarity threshold decides how similar a result must be to be returned."

Q48 How do you persist (save) a ChromaDB database to disk?

Use PersistentClient and provide a folder path.

ChromaDB automatically saves the data.

SHORT ANSWER

"Use PersistentClient. ChromaDB automatically saves the data to disk."

Q49 How do you load an existing ChromaDB collection after restarting the application?

Connect to the same PersistentClient and use get_collection().

SHORT ANSWER

"Connect to the same PersistentClient and use get_collection() to load the existing collection."

Q50 How do you store PDFs, Word documents, and text files in ChromaDB?

First, extract the text from the file.

Then, split it into chunks, create embeddings, and store the chunks in ChromaDB.

SHORT ANSWER

"I extract the text, split it into chunks, generate embeddings, and store them in ChromaDB."

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9010001847



info@vproskills.com



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